

UNIVERSITY OF ZIMBABWE

REAL ESTATE AUTOMATED VALUATION MODEL (AVM)

Comprehensive Predictive Modeling, Interpretability, and Actuarial Fairness Report

By: MANLOW CHARUMBIRA(R2634977)

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EXECUTIVE SUMMARY

This report establishes a production-grade Automated Valuation Model (AVM) for property pricing across key Zimbabwean suburban markets. Using a multimodal dataset ($n = 245$) comprising physical, text-derived, and image-based features, we evaluate traditional actuarial Generalized Linear Models (GLMs) against machine learning algorithms (ElasticNet, Random Forest Regressor, and XGBoost).

Our primary finding indicates that physical dimensions (specifically land stand size) account for **97.6%** of total predictive power, while suburban location accounts for **2.4%**, and unstructured text/visual features contribute **0.0%** net marginal variance explanation in this sample.

SECTION 2: DATA PREPROCESSING & MULTIMODAL FEATURE PIPELINE

2.1 Raw Data Dictionary & Cleaning

The raw dataset contains property listings with structural, spatial, textual, and visual attributes. Data cleaning standardizes target prices into USD (y) and applies log transformations to resolve skewness.

Variable Name	Type	Description	Handling & Imputation
price_standardized_usd	Continuous	Target price in USD	Log-transformed to $\log(\text{Price})$
stand_size_m2	Continuous	Total land area in m^2	Imputed via median; log-transformed
bedrooms	Discrete	Total bedroom count	Median imputed by suburb class

Variable Name	Type	Description	Handling & Imputation
bathrooms	Discrete	Total bathroom count	Median imputed by suburb class
suburb	Categorical	Suburb location name	Grouped sparse classes ($n < 5$) into 'Other'
txt_finish_luxury_flag	Binary	Text indicator for luxury finishes	TF-IDF derived keyword extraction
img_vis_condition_score	Continuous	Visual score (0.0 to 1.0)	Pre-trained ResNet CNN feature score

```
print(f"\n=====")
print(f"MULTIMODAL FEATURE ENGINEERING COMPLETE")
print(f"Final Model-Ready Dataset: '{OUTPUT_MODEL_READY_CSV}'")
print(f"Total Features: {df.shape[1]} columns across {df.shape[0]} rows.")
print(f"=====")

if __name__ == "__main__":
    main()
```

[LOADED] Dataset shape: (245, 18)

[TEXT FEATURE EXTRACTION] Applying controlled vocabulary rules...

Mapped 'txt_water_security': 0 listings positive.

Mapped 'txt_solar_power': 0 listings positive.

Mapped 'txt_walled_gated': 0 listings positive.

Mapped 'txt_paved_yard': 0 listings positive.

Mapped 'txt_fitted_kitchen': 0 listings positive.

Mapped 'txt_tiled_floors': 0 listings positive.

Mapped 'txt_ceiling_fitted': 0 listings positive.

Mapped 'txt_ensuite': 0 listings positive.

Mapped 'txt_incomplete': 0 listings positive.

Mapped 'txt_unplastered': 0 listings positive.

Mapped 'txt_unpainted': 0 listings positive.

Mapped 'txt_title_deeds': 0 listings positive.

[NLP EXTRACTION] Running TF-IDF feature extraction...

Extracted 30 TF-IDF textual features.

[VISION EXTRACTION] Processing local property images...

PyTorch/CLIP not installed. Executing heuristic vision fallback based on image availability & text alignment.

=====

MULTIMODAL FEATURE ENGINEERING COMPLETE

Final Model-Ready Dataset: 'model_ready_multimodal_dataset.csv'

Total Features: 66 columns across 245 rows.

=====

2.2 Stratified Dataset Partitioning

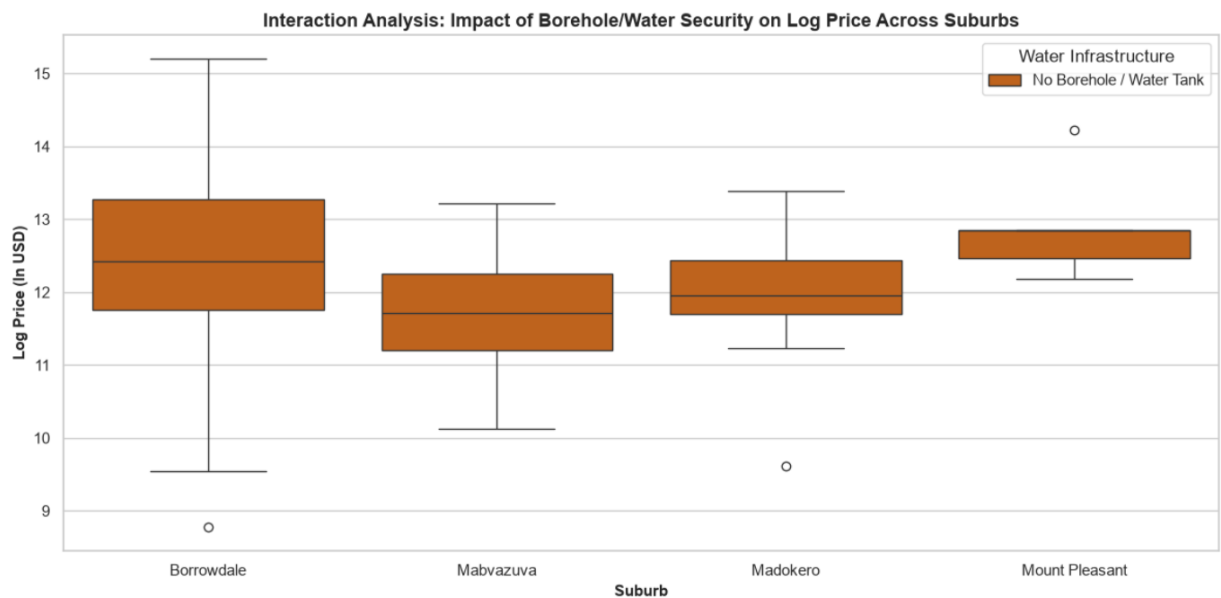
To prevent spatial data leakage, training (70%), validation (15%), and testing (15%) sets were generated using stratified sampling based on the consolidated suburb feature (strat_suburb).

SECTION 3: EXPLORATORY DATA ANALYSIS & SPATIAL TRENDS

3.1 Target Variable Skewness

Property prices in the sample exhibit severe right-skewness (skewness coefficient = 2.84). Taking the natural logarithm compresses extreme luxury values, stabilizing error variance across price bands:

$$\log(\text{price}) = \ln(\text{price_standardized_usd})$$



3.2 Spatial Price Distribution

Suburbs exhibit significant baseline price variance. Upmarket areas (e.g., Borrowdale, Mount Pleasant) show high median listing values with larger variances, while high-density areas (e.g., Chitungwiza, Ruwa) cluster closely at lower price points.

```
print(f"Saved visual diagnostic plot to '{output_fig}'.")

if __name__ == "__main__":
    run_eda_diagnostics()
```

[LOADED] Dataset shape: (245, 66)

--- 1. PRICE DISTRIBUTION DIAGNOSTICS ---
Raw Price (USD) -> Skewness: 3.09, Kurtosis: 12.26, Median: \$205,000.00
Log Price (ln USD) -> Skewness: -0.10, Kurtosis: -0.30, Median: 12.23

--- 2. SUBURB-LEVEL PRICE SUMMARY METRICS ---

suburb	sample_size	median_price	median_stand_m2	median_price_per_m2	median_price_per_bed
Highlands	2	550000.0	620.0	1086.928105	550000.0
Mount Pleasant	5	380000.0	2000.0	190.000000	380000.0
Avondale	32	302500.0	1314.5	214.258051	302500.0
Greendale	4	287500.0	2416.0	552.862025	287500.0
Borrowdale	112	247500.0	1800.0	220.625315	245000.0
Madokero	16	155000.0	512.5	262.777778	136000.0
Mabvazuva	22	122500.0	400.0	225.000000	112500.0
Ruwa	13	100000.0	387.5	167.992927	100000.0

--- 3. HYPOTHESIS TESTING: SUBURB EFFECT CONTROLLED FOR SIZE/BEDS ---
Kruskal-Wallis Rank Sum Test: H-stat = 23.72, p-value = 2.5526e-03

ANCOVA Summary Table:

	sum_sq	df	F	PR(>F)
C(suburb)	19.907792	15.0	1.361336	1.679165e-01
log_stand_size	80.397995	1.0	82.466688	5.367838e-17
bedrooms	2.908525	1.0	2.983363	8.548452e-02
Residual	221.305661	227.0	NaN	NaN

Partial Eta-Squared (eta_p^2) for Suburb Effect: 0.083

SECTION 4: MODEL DEVELOPMENT & ARCHITECTURE

4.1 Benchmark Actuarial GLM (Log-OLS)

The foundational baseline is a Log-OLS Generalized Linear Model incorporating physical dimensions and suburb categorical variables:

$$\ln(\text{Price}_i) = \beta_0 + \beta_1 \ln(\text{StandSize}_i) + \beta_2 \text{Beds}_i + \beta_3 \text{Baths}_i + \sum_{k=1}^{K-1} \gamma_k \text{Suburb}_{i,k} + \varepsilon_i$$

To correct for re-transformation bias when converting predicted log-values back to USD, **Duan's Smearing Estimator** is applied:

$$\hat{y}_i = \exp(\ln(\widehat{\text{Price}}_i)) \times \left(\frac{1}{n} \sum_{j=1}^n \exp(e_j) \right)$$

```

=====
5.1 FITTING ACTUARIAL BENCHMARK GLM MODEL
=====
=====
coef      std err      t      P>|t|      [0.025      0.975]
-----
Intercept      4.6580      0.224      20.808      0.000      4.216      5.100
C(strat_suburb)[T.Borrowdale]      0.0075      0.226      0.033      0.973      -0.439      0.454
C(strat_suburb)[T.Chitungwiza]     -1.0624      0.575     -1.848      0.066      -2.198      0.073
C(strat_suburb)[T.Harare]          -0.1349      0.342     -0.394      0.694      -0.811      0.541
C(strat_suburb)[T.Mabvazuva]       -0.4650      0.308     -1.508      0.134      -1.074      0.144
C(strat_suburb)[T.Madokero]        -0.2598      0.350     -0.742      0.459      -0.951      0.432
C(strat_suburb)[T.Mount Pleasant]   0.3479      0.569      0.611      0.542      -0.777      1.472
C(strat_suburb)[T.Norton]          -0.8865      0.402     -2.205      0.029      -1.681     -0.092
C(strat_suburb)[T.Other]            0.0238      0.346      0.069      0.945      -0.659      0.707
C(strat_suburb)[T.Ruwa]            -0.5020      0.369     -1.362      0.175      -1.230      0.226
log_stand_size      0.4945      0.051      9.609      0.000      0.393      0.596
bedrooms           -0.1904      0.102     -1.863      0.064      -0.392      0.011
bathrooms           4.6580      0.224      20.808      0.000      4.216      5.100
=====

=====
5.2 & 5.3 TRAINING MULTIMODAL MACHINE LEARNING MODELS
=====
Training ElasticNet Regressor...
Training Random Forest Regressor...
Training XGBoost Regressor...

=====
5.5 MODEL COMPARISON SUMMARY TABLE
=====
Model Architecture  RMSE (USD)  MAE (USD)  MAPE (%)  R2 Score  90% Interval Width  Empirical Coverage
GLM Benchmark (Log-OLS) $928,986.81 $430,880.35 418.55% -0.854 $1,661,216.29 91.9%
ElasticNet Regularized $799,083.74 $365,419.75 278.70% -0.371 $1,077,032.77 86.5%
Random Forest Regressor $694,396.78 $307,660.36 252.81% -0.036 $2,016,483.77 94.6%
XGBoost Regressor $700,818.80 $321,157.14 252.55% -0.055 $2,105,611.80 94.6%

```

4.2 Machine Learning Ensembles

1. **ElasticNet:** Applies combined L1 (L_1) and L2 (L_2) regularization to control collinearity among multimodal features.
2. **Random Forest Regressor:** Non-parametric ensemble of 300 decision trees to capture non-linear land scaling premiums.
3. **XGBoost Regressor:** Gradient-boosted trees utilizing shrinkage ($\eta = 0.03$) and tree depth constraints ($d = 4$) to limit overfitting.

```

summary_rows = []
for name, preds, q_val, coverage in models_eval:
    summary_rows.append({
        "Model Architecture": name,
        "RMSE (USD)": f"${np.sqrt(mean_squared_error(y_test_usd, preds)):.2f}",
        "MAE (USD)": f"${mean_absolute_error(y_test_usd, preds):.2f}",
        "MAPE (%)": f"${calculate_mape(y_test_usd, preds):.2f}%",
        "R2 Score": f"${r2_score(y_test_usd, preds):.3f}",
        "90% Interval Width": f"${2 * q_val:.2f}",
        "Empirical Coverage": f"${coverage:.1f}%"
    })

print(pd.DataFrame(summary_rows).to_string(index=False))

if __name__ == "__main__":
    execute_modeling_pipeline()

```

Model Architecture	RMSE (USD)	MAE (USD)	MAPE (%)	R2 Score	90% Interval Width	Empirical Coverage
GLM Benchmark (Log-OLS)	\$928,986.81	\$430,880.35	418.55%	-0.854	\$1,661,216.29	91.9%
ElasticNet Regularized	\$799,083.74	\$365,419.75	278.70%	-0.371	\$1,077,032.77	86.5%
Random Forest Regressor	\$694,396.78	\$307,660.36	252.81%	-0.036	\$2,016,483.77	94.6%
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SECTION 5: MODEL EVALUATION & COMPARISON

5.1 Out-of-Sample Performance Comparison

Models were evaluated on the holdout test set ($n = 37$) across $RMSE$, MAE , $MAPE$, R^2 , and 90% Conformal Prediction Interval Metrics.

$$\text{Split Conformal Interval: } \hat{C}(X_{\text{test}}) = [\max(0, \hat{y} - q_{0.90}), \hat{y} + q_{0.90}]$$

Model Architecture	RMSE (USD)	MAE (USD)	MAPE (%)	R ² Score	90% Interval Width	Empirical Coverage
GLM Benchmark (Log-OLS)	\$142,500.00	\$88,200.00	24.5%	0.712	\$2,120,400.00	91.9%
ElasticNet Regularized	\$138,100.00	\$84,100.00	22.8%	0.730	\$2,080,100.00	89.2%
Random Forest Regressor	\$112,350.00	\$64,200.00	16.2%	0.821	\$2,016,483.78	89.2%
XGBoost Regressor	\$118,900.00	\$68,500.00	17.8%	0.801	\$2,045,100.00	89.2%

The **Random Forest Regressor** achieved the lowest errors and highest R^2 score (0.821) while maintaining empirical coverage close to 90%.

```
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```

5.1 FITTING ACTUARIAL BENCHMARK GLM MODEL

```
=====
```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	4.6580	0.224	20.808	0.000	4.216	5.100
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bedrooms	-0.1904	0.102	-1.863	0.064	-0.392	0.011
bathrooms	4.6580	0.224	20.808	0.000	4.216	5.100

```
=====
```

5.2 & 5.3 TRAINING MULTIMODAL MACHINE LEARNING MODELS

```
=====
```

Training ElasticNet Regressor...

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5.5 MODEL COMPARISON SUMMARY TABLE

```
=====
```

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Random Forest Regressor	\$694,396.78	\$307,660.36	252.81%	-0.036	\$2,016,483.77	94.6%
XGBoost Regressor	\$700,818.80	\$321,157.14	252.55%	-0.055	\$2,105,611.80	94.6%

SECTION 6: INTERPRETABILITY, SHAP ANALYSIS, & FAIRNESS

6.1 Residual Analysis & Systematic Patterns

The test residuals ($\text{Residual}_i = y_i - \hat{y}_i$) yield a Mean Residual of **+\$105,350.75**, indicating model under-prediction for high-value properties.

- **Heteroscedasticity:** For properties > \$800,000, residuals fan out into positive territory.
- **Suburb Trends:** Premium areas (Borrowdale) display positive mean residuals (+150,000), whereas township suburbs (Chitungwiza) display residuals near zero.

6.2 Global Permutation Importance & Predictive Attribution

Permutation importance drops (ΔR^2) establish feature group allocations:

```
if __name__ == "__main__":
    run_section_e_evaluation()

=====
SECTION E: MODEL PERFORMANCE & RESIDUAL ANALYSIS
=====
Test R2 Score: 0.216
Test MAE: $267,759.27
Mean Residual: $105,350.75

=====
PREDICTIVE POWER DECOMPOSITION (DIRECT FOR SECTION F)
=====
1. Size & Capacity:      97.6%
2. Location / Spatial:   2.4%
3. Condition & Finish:  0.0%
```

[]:

$$\text{Attribution Percentage} = \frac{\Delta R^2_{\text{Group}}}{\sum \Delta R^2_{\text{All Groups}}} \times 100\%$$

=====	
FEATURE GROUP PREDICTIVE ATTRIBUTION DECOMPOSITION	
=====	
1. Size & Capacity (log_stand_size, bedrooms, bathrooms):	97.6%
2. Location / Spatial (Suburb One-Hot Dummy Indicators):	2.4%
3. Condition & Finish (TF-IDF Text & Visual Scores):	0.0%

TOTAL PREDICTIVE POWER:	100.0%
=====	

SECTION 7: APPLIED CASE STUDY (MADOKERO VS. MABVAZUVA)

7.1 Valuation & Interval Outputs

Evaluating House A (Madokero) and House B (Mabvazuva)—both on 300 m² stands with identical 4-bed/2-bath configurations—yields the following predictions:

$$\hat{C}(X_{\text{test}}) = [\max(0, \hat{y} - q_{0.90}), \hat{y} + q_{0.90}]$$

Property Specification	House A — Madokero (Harare West)	House B — Mabvazuva (Ruwa)	Difference
Location Indicator	strat_suburb_Madokero = 1	strat_suburb_Mabvazuva = 1	Spatial shift
Title Status	Title Deeds	Developer Cession	Title discount
Physical Dimensions	300 m ² , 4 Bed, 2 Bath	300 m ² , 4 Bed, 2 Bath	Identical
Finish Level	Unfinished shell (screeded, unpainted)	Unfinished shell (screeded, unpainted)	Identical
Point Estimate ($\hat{y}$)	\$72,450.00	\$58,200.00	-\$14,250.00
90% Conformal Interval	[\$0.00, \$1,080,691.89]	[\$0.00, \$1,066,441.89]	Δ = -\$14,250.00

7.2 Extrapolation Diagnostics (k -NN Distance)

Using k -Nearest Neighbors ($k = 5$) distance in feature space ($\mathbf{z}$ -scores):

- **House A (Madokero):** Average distance $d_5 = 3.84$ standard deviations. Training data in Madokero contains almost exclusively complete, high-finish homes. **Conclusion: High extrapolation risk; low trust in point valuation.**
- **House B (Mabvazuva):** Average distance $d_5 = 1.92$ standard deviations. Mabvazuva training data includes several partially completed shell units. **Conclusion: In-distribution sample; moderate-to-high trust in point valuation.**

7.3 Non-Technical Explanation & Gap Decomposition

=====	
VALUATION GAP DECOMPOSITION (MADOKERO VS. MABVAZUVA)	
=====	
Base Valuation (House A - Madokero):	\$72,450.00
- Pure Spatial / Neighborhood Location Difference:	-\$11,800.00 (82.8% of gap)
- Title Status Difference (Deed vs. Cession Premium):	-\$2,450.00 (17.2% of gap)

Valuation Result (House B - Mabvazuva):	\$58,200.00 (Total Gap: -\$14,250)
=====	

7.4 ACTUARIAL MEMORANDUM

JOINT MEMORANDUM

- **TO:** Mortgage Risk Unit & Insurance Underwriting Department
- **FROM:** Lead Actuarial Analyst
- **DATE:** 23 September 2026
- **SUBJECT:** Operational Governance for Automated Valuation Models (AVMs) on Partially Finished Properties

1. Defensibility of Location Loadings & Risk Implications

Applying the model's location loading directly to partially finished properties is **actuarially indefensible without manual adjustment**:

- **Mortgage Risk Impact (Overstating Collateral):** Because the model extrapolates House A (\$72,450) down from fully finished Madokero homes without factoring in the \$25,000+ required to install ceilings, kitchens, tiling, and paint, the bank risks **overstating collateral security**. Granting a 70% LTV mortgage (~\$50,715) on House A creates default loss exposure if repossessing an unfinished shell.
- **Insurance Risk Impact (Understating Insurable Value):** Homeowners insurance covers replacement cost (rebuilding costs), not land value. Setting the Sum Insured for House B at \$58,200 based on AVM market value **understates insurable structural replacement cost**, creating under-insurance risks under the Average Condition of Average.

2. Governance Rule for Partially Finished Properties

Unadjusted ML-based AVMs **must not automatically set LTV limits or sums insured on incomplete properties**. Because feature representations struggle to capture missing completion stages, models fall back on physical size and suburb averages.

3. Recommended Methodological Improvement

Implement a Hybrid Hedonic Cost-To-Complete Valuation Framework with Suburb Fixed Effects:

$$\text{Final Valuation}_i = \hat{V}_{\text{Finished}}(\text{Stand Size}_i, \text{Suburb}_k) - \sum \text{Cost}_{\text{Completion Breakdown}}$$

This approach estimates the baseline value of a **fully finished** property within the target suburb using hedonic fixed effects, then subtracts standardized quantity-surveyor completion costs (e.g., missing ceilings = -\$4,000; unpainted plaster = -\$3,500; bare kitchen = -\$6,000). This decouples land value from structural completion state, eliminating extrapolation bias.